NEP-2020 Readiness Index Construct Design – WIP (myelin internal brainstorming)

A. Weight distribution across Readiness Constructs (Total = 100)

This weighting reflects **"what unlocks change"**, not "what looks good early".

Readiness Construct	Weight	Why this weight (construct logic)
Intent Readiness	**20**	Necessary precondition. Without intent, nothing moves-but intent alone is cheap and over-reported.
Practice Readiness	**25**	Core enactment layer. This is where NEP intent becomes classroom reality through depth + reflection.
Capacity Readiness	**20**	Teacher cognition is the limiting reagent for diagnostic and adaptive practice.
System Readiness	**20**	Systems determine whether good practice sustains or collapses under load.
Ecosystem Readiness	**15**	Validation and early impact matter, but are downstream and sample-based in SRI v1.
Total	**100**	Balanced, non-dominant, trajectory-aware

Key principle respected:
No single construct exceeds 25 → avoids domination and false precision.

B. Mapping: 7 Index Signals → Construct → Weight → Value in Index

Map each signal explicitly, without early aggregation (Layer 1 logic preserved).

A. Intent Readiness (20)

Signal	Weight	Value it contributes
Leader Orientation (FP1–FP5)	**10**	Sets priority climate, legitimises NEP-aligned risk-taking.
Teacher Orientation (FP1–FP5)	**10**	Reveals where teachers naturally place attention before behaviour changes.

Why split evenly:
Misalignment (leader high, teacher low or vice versa) is itself a diagnostic signal, not noise.

B. Practice Readiness (25)

Signal	Weight	Value it contributes
Practice Depth (D1–D4 across FPs)	**25**	Primary evidence of enactment maturity (procedural → adaptive).

Why this is the heaviest construct:
NEP-2020 lives or dies here. Depth + reflection is where "learn to learn" conditions emerge.

C. Capacity Readiness (20)

Signal	Weight	Value it contributes
Teacher Cognitive Capacity (Pre/Post)	**20**	Determines ceiling for diagnostic reasoning, abstraction, and adaptation.

Why single-signal but high weight:
Cognition is not inferred- it is directly measured and strongly predictive of depth progression.

D. System Readiness (20)

Signal	Weight	Value it contributes
Teacher Enablement (Experience lens)	**10**	Whether systems reduce friction or add cognitive load.
Leadership Routines (Structure lens)	**10**	Whether practice is protected, reviewed, and normalised institutionally.

Important nuance:
Strong systems with weak practice ≠ readiness
Strong practice with weak systems = fragile readiness

SRI surfaces this tension deliberately.

E. Ecosystem Readiness (15)

Signal	Weight	Value it contributes
Parent Validation Signals	**7**	Confirms whether learning communication and trust are felt externally.
Early Student Signals (sample-based)	**8**	Indicates exposure to cognitive demand, not achievement.

Why slightly more weight to student than parent:
Students reflect classroom conditions more directly; parents validate alignment and credibility.

1. Intent Readiness (20)

Purpose

Intent Readiness captures whether a school's leadership and teachers demonstrate a shared, structurally complete orientation toward NEP-2020, sufficient to support meaningful implementation in practice.

Intent Readiness **does not** assess behaviour or outcomes.
It surfaces where attention naturally goes when leaders and teachers think about teaching, learning, and school priorities.

Signals Contributing to Intent Readiness

Intent Readiness is derived from **two independent raw signals**:

1. **Leader FP Orientation**
2. **Teacher FP Orientation**

Each signal is analysed separately and then synthesised to produce the construct value.

Maximum construct weight: 20

- Leader Intent: 10
- Teacher Intent: 10

Raw signal definition

FP Orientation Responses

Respondents (leaders or teachers) answer a structured set of questions, where each response option is mapped to one of the five NEP-2020 Foundational Principles (FPs):

- FP1: Every Child is Unique

- FP2: Holistic, Experiential Learning
- FP3: Teacher as Reflective Practitioner
- FP4: Assessment for Learning
- FP5: Collaboration & Community

Each selection is treated as an **attention signal**, not a correctness judgment.

Step 1: FP-wise Attention Density Vector (ADV)

Definition

For a given group (leaders or teachers), responses are converted into an **FP-wise Attention Density Vector**:

ADV=[d1,d2,d3,d4,d5]

where:

di=Number of selections mapped to FPi / Total selections

Properties:

- $\sum di=1$
- Each $di \in [0,1]$

Why this is used

- Removes sample size effects
- Preserves **relative emphasis** across principles
- Avoids over-rewarding volume of responses

The ADV is the **base representation** of intent.

Example: Teacher FP Attention Density (ADV)

Assume 10 questions × 20 teachers = 200 selections.

FP	Count	Density
FP1	60	0.30
FP2	40	0.20
FP3	70	0.35
FP4	20	0.10
FP5	10	0.05
Total	200	**1.00**

So teacher ADV =
[0.30, 0.20, 0.35, 0.10, 0.05]

This vector is **the signal**.

Step 2: Breadth Component (Coverage)

Definition

Breadth measures whether intent meaningfully spans NEP principles, or is narrowly concentrated.

A minimum attention threshold τ is defined (default: 0.10).

τ may be adjusted during sensitivity testing, but is fixed within a given SRI cycle.

Coverage=Number of FPs where $d_i \geq \tau / 5$

Range: [0,1]

Interpretation

- High coverage → multiple principles receive meaningful attention
- Low coverage → blind spots in NEP intent

Coverage protects against **partial NEP adoption**.

Step 3: Coherence Component (Balance)

Definition

Coherence measures whether NEP intent is **structurally complete enough to survive real classroom complexity**, rather than collapsing into one dominant belief.

Balance does not imply equal emphasis across principles; it indicates that no principle is systematically neglected.

Coherence is operationalised as the **absence of extreme skew** in the Attention Density Vector (**ADV**)

Computation

The maximum variance (0.16) is a theoretical bound determined by the number of Foundational Principles (5), corresponding to complete concentration of intent in a single principle. It is not derived from empirical data.

1. Reference distribution (structural completeness baseline):

U=[0.2,0.2,0.2,0.2,0.2]

2. Compute population variance of ADV:

$$Var = \frac{1}{5}\sum_{i=1}^{5}(di - 0.2)^2$$

Example

FP	di	di − 0.20	(di − 0.20)²
FP1	0.30	+0.10	0.0100
FP2	0.20	0.00	0.0000
FP3	0.35	+0.15	0.0225
FP4	0.10	−0.10	0.0100
FP5	0.05	−0.15	0.0225

Sum			**0.0650**

Variance = 0.0650 / 5 = 0.013

3. Compute maximum possible variance (all attention on one FP):

   Maximum variance occurs when **all attention collapses into one FP**:

[1.0, 0, 0, 0, 0]

$$Var_{max} = 0.16$$

Example

Compute variance:

FP	di	di − 0.20	$(di - 0.20)^2$
FP1	1.00	+0.80	0.6400
FP2	0.00	−0.20	0.0400
FP3	0.00	−0.20	0.0400
FP4	0.00	−0.20	0.0400
FP5	0.00	−0.20	0.0400
Sum			**0.8000**

Max Variance = 0.8000 / 5 = 0.16 (this is worst case tunnel vision)

4. Normalise variance:

$$Var_{norm}=Var/Var_{max}$$

   Example

   Normalised Variance = 0.013 / 0.16 ≈ 0.081

5. Convert to Balance score:

$$\text{Balance} = 1 - \text{Var}_{\text{norm}}$$

Range: [0,1]

Interpretation

- Balance ≈ 1 → focused yet structurally complete intent
- Balance → 0 → tunnel vision

Coherence protects against **ideological monoculture**, even when alignment exists.

Step 4: Alignment Component (Leader–Teacher Alignment)

Definition

Alignment measures whether leaders and teachers are oriented toward **the same NEP principles**, even if their emphasis differs. It measures directional similarity of attention, not agreement on intensity, quality, or practice.

Given:

- ADV_L = Leader Attention Density Vector
- ADV_T = Teacher Attention Density Vector

Alignment is computed using cosine similarity:

$$\text{Alignment} = ADV_L \cdot ADVT / \|ADV_L\| \, \|ADV_T\|$$

Range: [0,1]

Cosine similarity answers: "If we ignore scale, how similarly are attention weights distributed?"

Alignment is symmetric; leader–teacher and teacher–leader comparisons yield the same value.

Example

Leader ADV (L): = [0.25, 0.25, 0.20, 0.20, 0.10]

Teacher ADV (T): = [0.30, 0.20, 0.35, 0.10, 0.05]

Dot product (shared emphasis)

L·T=(0.25×0.30)+(0.25×0.20)+(0.20×0.35)+(0.20×0.10)+(0.10×0.05)

0.075 + 0.050 + 0.070 + 0.020 + 0.005 = 0.220

Vector magnitudes:

$$\|L\| = \sqrt{0.25^2 + 0.25^2 + 0.20^2 + 0.20^2 + 0.10^2} = \sqrt{0.215} = 0.464$$

$$\|T\| = \sqrt{0.30^2 + 0.20^2 + 0.35^2 + 0.10^2 + 0.05^2} = \sqrt{0.265} = 0.515$$

Cosine similarity (Alignment)

Alignment= L·T/ $\|L\| \times \|T\| = 0.220/0.464 \text{ X } 0.515 = 0.92$

Interpretation

- Alignment ≈ 1 → shared intent direction
- Low alignment → fragmentation between leadership and classroom priorities

Alignment protects against **disconnect between vision and practice**.

Step 5: Intent Signal Contribution (per group)

For leaders and teachers separately, an **Intent Signal Score** is computed as:

IntentScore = Coverage × Balance × Alignment

This multiplicative structure ensures that:

- No single strength can compensate for a structural weakness
- Narrow, skewed, or misaligned intent is penalised

Step 6: Construct Aggregation

- Leader Intent Score is mapped to **0–10**
- Teacher Intent Score is mapped to **0–10**

Intent Readiness=Leader Intent + Teacher Intent

Maximum = 20

An increase in Intent Readiness represents

If Intent Readiness increases, it must correspond to observable changes such as:

- Attention spreading to previously neglected NEP principles
- Reduction in single-principle dominance
- Greater convergence between leadership and teacher priorities

If such changes cannot be articulated, the computation is considered invalid.

Design Guardrails (explicit)

- Intent Readiness does not imply implementation quality
- High intent with low practice is a valid and expected state
- Alignment without coherence is treated as constrained readiness

2. Practice Readiness (25)

Step 1: Practice Depth Distribution Matrix (PDDM)

Purpose of Step 1

The Practice Depth Distribution Matrix captures how NEP intent is enacted in classrooms, by jointly representing:

- Which Foundational Principles (FP1–FP5) are being enacted
- At what depth (D1–D4) those practices operate

This step does not judge quality or effectiveness.
It establishes a structural map of classroom practice.

Raw Signal Definition (Input)

Practice Depth Responses

Teachers respond to classroom-practice items where each response is mapped to:

- One Foundational Principle (FP1–FP5)
- One Practice Depth band (D1–D4)
- Design exclusion FP1-D1, FP1-D4 and FP2-D4 (non-observable)

Practice Depth Bands

Depth	Meaning (operational)
D1	Procedural / rule-following
D2	Responsive / situational adjustment
D3	Diagnostic / evidence-based decision
D4	Adaptive / intentional redesign

Each response represents **how practice is enacted**, not whether it is correct or desirable.

Step 1.1: Constructing the Matrix

For a given school (or teacher cohort), responses are aggregated into a **5 × 4 matrix**:

PDDM=

$$\begin{matrix} FP1D1 & FP1D2 & FP1D2 \vdots\vdots \\ FP2D1 & FP2D2 & FP2D2 \vdots\vdots \\ FP3D1 & FP3D2 & FP3D3 \vdots\vdots \end{matrix}$$

Where each cell contains the **count of responses** mapped to that FP–Depth combination.

Step A1: Start from the normalised PDDM

Let pi, jp be the normalised cell value for FP i∈{1..5} and depth j∈{1..4} such that:

- pi,j ∈ [0,1]
- $\sum_{i=1}^{5} \sum_{j=1}^{4} p_{i,j=1}$

Step A2: Add an "observability mask"

Define $M_{i,j} \in \{0,1\}$

$Mi, j \in \{0,1\}$, where:

- $Mi, j = 1$: this FP–Depth is **observable** (instrument has items for it)
- $Mi, j = 0$: **not observable by design**

Known exclusions:

- MFP1,D1=0
- MFP1,D4=0
- MFP2,D4=0

Everything else is 1.

This ensures we **do not treat those cells as "zero practice."** They are simply **not measured**.

Step A3: Convert to FP-conditional depth distributions (only over observable depths)

First compute FP row totals over observable depths:

$$\mathrm{ri} = \sum_{\mathrm{j:Mij=1}} \mathrm{Pi,j}$$

Then define the within-FP depth distribution:

$$q_{i,j} = \begin{cases} \frac{p_{i,j}}{r_i} & \text{if } M_{i,j} = 1 \text{ and } r_i > 0 \\ 0 & \text{otherwise} \end{cases}$$

Now $\sum_{j:M_{i,j}=1} q_{i,j} = 1$ for every FP with evidence.

Step A4: Define depth weights (0–1 scale)

Use a simple monotonic progression:

- wD1=0
- wD2=0.33
- wD3=0.67
- wD4=1.00

(Any monotonic set is acceptable; this is clean and interpretable.)

Step A5: Compute FP-wise expected depth, then re-normalise for observability

Raw expected depth for FP iii:

$$E_i = \sum_{j:M_{i,j}=1} q_{i,j}\, w_j$$

Because some FPs cannot "reach" D4 (FP2) or cannot express D1/D4 (FP1), we must normalise by the **achievable depth range** for that FP:

Let:

$$w_i^{min} = \min_{j:M_{i,j}=1}(w_j), \quad w_i^{max} = \max_{j:M_{i,j}=1}(w_j)$$

Then:

$$E_i^{norm} = \begin{cases} \frac{E_i - w_i^{min}}{w_i^{max} - w_i^{min}} & \text{if } w_i^{max} > w_i^{min} \\ 0.5 & \text{if only one depth is observable (rare)} \end{cases}$$

This is the exact equivalent of "0.16 max variance bound" idea in Intent**- we're bounding each FP's depth score by what is *measurable*, not what is philosophically possible.**

Step A6: Add FP enactment coverage (prevents "depth in one FP only")

Define an FP evidence threshold τ

$$Coverage_{FP} = \frac{\#\{i : r_i \geq \tau_{fp}\}}{5}$$

Step A7: Aggregate to a single Depth Effectiveness (0–1)

We avoid row-total dominance by using **equal FP weighting** across those with evidence:

$$DepthEffectiveness = Coverage_{FP} \times \frac{\sum_{i:r_i>0} E_i^{norm}}{\#\{i : r_i > 0\}}$$

Step A8: Map to Depth score (0–25)

$DepthScore_{0\text{-}25} = 25 \times DepthEffectiveness$

Worked Example (Illustrative)

Given (Instrument observability constraints)

Non-observable cells (by design):

- FP1–D1 = NA
- FP1–D4 = NA
- FP2–D4 = NA

So these cells are **not captured** and must be **excluded from totals**.

1) Observable raw counts (example)

Let the observed counts be:

FP \ Depth	D1	D2	D3	D4	Row Total (Observed)
FP1	**NA**	10	6	**NA**	16
FP2	14	12	6	**NA**	32
FP3	20	18	12	5	55
FP4	18	16	8	3	45
FP5	15	10	6	4	35
Total observed $N_{o\beta s}$					**183**

For computation, we use $\mathbf{N_{o\beta s}}$ = 183.

2) Normalised observed PDDM (p_{ij})

Divide each observed cell by **183**.

Example cells:

- FP1–D2 = 10/183 = **0.0546**
- FP1–D3 = 6/183 = **0.0328**
- FP2–D1 = 14/183 = **0.0765**
- FP3–D4 = 5/183 = **0.0273**

Row totals (over observable depths) are:

- r_1 = 16/183 = **0.0874**
- r_2 = 32/183 = **0.1749**
- r_3 = 55/183 = **0.3005**
- r_4 = 45/183 = **0.2459**
- r_5 = 35/183 = **0.1913**

3) Within-FP depth distributions (q_{ij})

These are computed **within each FP using only observable depths**:

FP1 (only D2, D3 observable)

- q_1,D2 = 10/16 = **0.625**
- q_1,D3 = 6/16 = **0.375**

FP2 (D1–D3 observable)

- q_2,D1 = 14/32 = **0.4375**
- q_2,D2 = 12/32 = **0.375**
- q_2,D3 = 6/32 = **0.1875**

4) Depth weights (example set)

- D1 = 0
- D2 = 0.33
- D3 = 0.67
- D4 = 1.00

5) FP-wise expected depth (E_i)

FP1

E_1 = (0.625×0.33) + (0.375×0.67)
= 0.206 + 0.251
= **0.457**

But FP1 only has D2–D3 observable ⇒ achievable range is [0.33, 0.67].
So we normalise by achievable range:

E_1^n = (0.457 - 0.33) / (0.67 - 0.33)
= 0.127 / 0.34
= **0.374**

FP2

E_2 = (0.4375×0) + (0.375×0.33) + (0.1875×0.67)
= 0 + 0.124 + 0.126
= **0.25**

Achievable range is [0, 0.67] (D1–D3 only).
E_2^n = 0.25 / 0.67 = **0.373**

FP3, FP4, FP5 (full observable range)

- E_3^n = **0.345**
- E_4^n = **0.303**
- E_5^n = **0.323**

6) FP Coverage (must be observability-aware)

If you use a flat threshold like τ_fp = 0.10 on r_i, FP1 will falsely fail (r_1 = 0.0874) **only because FP1 has fewer observable cells**.

So define expected FP share based on observability design:

Observable depth-counts per FP:

- FP1: 2 depths
- FP2: 3 depths
- FP3: 4
- FP4: 4
- FP5: 4
  Total = 17

Expected share:

- e_1 = 2/17 = 0.1176
- e_2 = 3/17 = 0.1765

- $e_3 = e_4 = e_5 = 4/17 = 0.2353$

Now compute evidence ratio:

- $ratio_i = r_i / e_i$

$$Coverage_{FP} = \frac{\#\{i : ratio_i \geq 0.5\}}{5}$$

If we set a simple rule like **$ratio_i \geq 0.5$**, all FPs pass here ⇒ Coverage_FP = **1.0**

7) DepthEffectiveness (0–1)

Mean of normalised FP depths:

(0.374 + 0.373 + 0.345 + 0.303 + 0.323) / 5
= 1.718 / 5
= **0.344**

DepthEffectiveness = Coverage_FP × mean = 1.0 × 0.344 = **0.344**

8) Map to Depth Score (0–25)

DepthScore = 25 × 0.344 = **8.6/25**

Why this matrix is necessary and cannot be skipped

PDDM preserves

- Depth distribution **(D1 vs D3 is visible)**
- Principle-wise enactment **(which FPs stay procedural)**
- Structural gaps **(e.g., FP4 never reaches D3/D4)**

What it deliberately avoids

- No averaging across depths
- No collapsing FP differences
- No premature "score"

This is the **practice analogue** of the Attention Density Vector in Intent Readiness.

Conceptual Interpretation

From the matrix alone, we can already observe:

- Practices concentrated in **D1/D2** → enactment without diagnosis
- Sparse **D3/D4** cells → limited learner-responsive practice
- Uneven FP rows → partial NEP enactment

But **no judgement is made yet**.

Design Guardrails

- The matrix is **descriptive, not evaluative**
- High D1 presence is expected in early stages
- D4 scarcity is normal and not penalised here
- Interpretation may use Trajectory signals analytically, but Trajectory is not included in Practice Readiness scoring in SRI v1.

3. Capacity Readiness (20)

Purpose

Capacity Readiness measures whether teachers possess the **cognitive capability required to adopt, interpret, and enact NEP-aligned practice at increasing depth.**

Capacity represents the **internal ceiling** of change.

While Intent indicates orientation and Practice indicates enactment, Capacity indicates whether teachers have the **reasoning capability necessary for diagnostic and adaptive teaching.**

Capacity is treated as a **structural readiness condition**, not a behavioural compliance indicator.

What Capacity Is (v1 Scope)

Capacity Readiness captures:

- Ability to engage in higher-order thinking (Apply, Analyse, Evaluate)
- Ability to decompose problems
- Ability to abstract from specific cases
- Willingness to revise reasoning based on evidence
- Stability of reasoning patterns under uncertainty

Capacity does not include:

- Login behaviour
- Reflection frequency
- System enablement
- Practice depth (measured separately)
- Short-term intervention participation

Signal Contributing to Capacity Readiness

Capacity Readiness is derived from a Cognitive Pattern Index (CPI) computed from structured cognitive probes used in the ecosystem (e.g., MathTangle-style diagnostic tasks).

Maximum construct weight: 20 No sub-weighting is introduced in v1.

Step 1: Cognitive Probe Signals

Each teacher completes a structured probe that captures:

1. Selected option(s)
2. Time spent per item
3. Sequence of interaction
4. Whether answer was revised
5. Completion vs skip behaviour

These are treated as **pattern signals**, not correctness scores.

Step 2: Deriving Cognitive Pattern Dimensions (CPDs)

From probe interaction data, derive the following dimensions:

CPD	Description
CPD1	Higher-order engagement (Analyse/Evaluate proportion)
CPD2	Decomposition / abstraction behaviour
CPD3	Revision behaviour under uncertainty
CPD4	Consistency of reasoning across items

Each CPD is normalised to a 0–1 range.

Step 3: Compute Cognitive Pattern Index (CPI)

Let:

$$CPD_k \in [0,1] \quad \text{for } k = 1..4$$

Define CPI as:

$$CPI = \frac{CPD_1 + CPD_2 + CPD_3 + CPD_4}{4}$$

Range:

$$CPI \in [0,1]$$

Range: CPI∈[0,1]

This ensures:

- No single dimension dominates
- Score remains bounded
- Computation remains transparent

Step 4: Map to Capacity Readiness (0–20)

CapacityReadiness0-20=20×CPI

Interpretation

Higher Capacity Readiness indicates:

- Greater cognitive ceiling for diagnostic teaching
- Greater potential to sustain D3/D4 practice
- Higher likelihood of benefiting from advanced pedagogical interventions

Lower Capacity Readiness indicates:

- Cognitive constraints limiting depth progression
- Need for foundational reasoning support before practice deepening

Design Guardrails

1. Capacity Readiness is independent of Practice Depth.
2. A teacher cohort may show:
    - High capacity but low practice (blocked by system)
    - Low capacity but moderate practice (ritualised compliance)
3. Capacity does not fluctuate with short-term participation.
4. Pre–post cognitive shifts may be analysed separately but are not embedded in v1 scoring.

4. System Readiness (20)

Purpose

System Readiness measures whether the school's **operational academic leadership structure** actively sustains and protects NEP-aligned practice.

While Intent reflects belief and Practice reflects enactment, System Readiness reflects:

Whether institutional structures reduce friction and protect diagnostic teaching under pressure.

System Readiness is fully **top-down actionable** and can be strengthened through structured intervention with minimal manual overhead.

Respondents

System Readiness is assessed through a structured audit completed by the:

- Principal
- Vice Principal(s)
- Academic Coordinator / Lead Teacher

Trustees or non-operational management members are not respondents to this construct.

Signals Contributing (20)

Sub-Construct	Weight	Lens
Teacher Enablement	10	Experience lens
Leadership Routines	10	Structural lens

Balanced intentionally:

- One reflects teacher operating conditions.
- One reflects institutionalised review architecture.

A. Teacher Enablement (0–10)

Definition

Teacher Enablement measures whether institutional conditions reduce cognitive and administrative friction and provide structural support for deep instructional practice. It answers does the system make it easier or harder for teachers to operate at D2–D4 levels?

Enablement Indicators (Binary Structural)

Each condition is scored:

- 1 = Exists and structurally embedded
- 0 = Absent or ad-hoc

Code	Structural Condition
E1	Protected planning time exists in timetable
E2	Regular structured academic review meetings occur
E3	Assessment/data access is available to teachers
E4	Administrative load does not displace instructional time
E5	Leadership formally encourages experimentation

Computation

EnablementIndex = (E1+E2+E3+E4+E5)/ 5

TeacherEnablement0-10 = 10 × EnablementIndex

Range: 0–10

B. Leadership Routines (0–10)

Definition

Leadership Routines measure whether NEP-aligned instructional practice is structurally reviewed, monitored, and institutionalised. It answers the practice of diagnostic/adaptive teaching expected and structurally reviewed.

Routine Indicators (Binary Structural)

Each scored:

- 1 = Exists and consistently implemented
- 0 = Absent or informal

Code	Structural Routine
R1	Classroom observation framework aligned to D1–D4 exists
R2	Observation calendar is structured and scheduled
R3	Feedback template is standardised and documented
R4	Intervention tracking mechanism exists
R5	Principal-level academic review cadence exists

Computation

RoutineIndex=R1+R2+R3+R4+R5/5

LeadershipRoutines0-10 = 10 × RoutineIndex

Range: 0–10

C. Aggregate System Readiness (0–20)

SystemReadiness0-20 = TeacherEnablement0-10 + LeadershipRoutines0-10

Maximum = 20

Interpretation

High System Readiness

- Practice depth can sustain growth.
- Capacity can be leveraged.
- Intervention impact is scalable.

Low System Readiness

- Even strong teachers struggle to sustain D3/D4.
- Practice remains fragile.
- Improvement efforts decay without reinforcement.

5. Ecosystem Readiness (15)

Purpose

Ecosystem Readiness captures whether NEP-aligned intent, practice, capacity, and system shifts are beginning to be:

- Felt externally (by parents)
- Reflected in student exposure to higher-order cognitive demand

Ecosystem Readiness does not measure final outcomes. It measures early validation and directional alignment. It answers the translation into visible signals beyond internal system from structural shifts.

All sub-signals are normalised to a 0–1 index before weight application, ensuring measurement scale consistency across binary and Likert-based instruments.

Signals Contributing (15)

Sub-Signal	Weight	Role

Parent Validation Signals	7	External trust and communication alignment
Early Student Cognitive Signals (sample-based)	8	Exposure to higher-order demand

Slightly higher weight to student because students reflect classroom exposure more directly than parent perception.

A. Parent Validation Signals (0–7)

It does not measures

satisfaction.
branding.
reputation.

It measures whether parents experience:

- Clear learning communication
- Transparency in assessment approach
- Trust in teacher intent
- Perception of cognitive challenge

Design Principle

Parent signals are perception-based but structured.

Binary or scaled (0–1) structural questions, such as:

- P1: Parent understands learning goals beyond marks
- P2: Parent receives structured feedback communication
- P3: Parent observes cognitive challenge in student work
- P4: Parent trusts teacher judgement
- P5: Parent-school communication loop exists

Let:

ParentIndex = P1+P2+P3+P4+P5/5

ParentReadiness0-7 = 7 × ParentIndex

B. Student Cognitive Exposure Signals (0–8)

This is the sensitive part. Cognitive and competency baseline signals for small sample of student at teacher.

Definition

Student Cognitive Exposure measures whether students are being exposed to:

- Apply-level tasks
- Analyse-level prompts
- Abstraction and decomposition
- Revision opportunities

It does not measure correctness or achievement.

Step 1: Representative Sample Selection

Sample must:

- Cover multiple grades
- Include teachers across capacity bands
- Represent a minimum coverage threshold (e.g., ≥30% of cohort)

Sample selection approach

Use two layers, but only one affects the SRI score:

1. **Index Sample** (counts toward StudentReadiness 0–8):
   Use your representative sample (at least three/four teacher in a school covering all grades (STD 3-8), but make it stratified, not purely random:
   - Ensure coverage across schools + grades.
   - Include teachers across capacity bands (explicitly required).
   - Apply CoverageFactor to prevent inflation.
2. Validation Sample (does NOT affect SRI score):
   Use high-cognitive teacher sample as a "transfer / mechanism" lens:
   - Report it as a *separate analytic* (like a robustness check).
   - Use to interpret gaps:
     - High internal readiness + low ecosystem = "external lag / early-stage shift" (already in your interpretation grid).
     - High capacity but low student exposure in best classrooms → points to System/Practice constraints.

This keeps SRI honest while still extracting the learning.

Step 2: Compute ExposureIndex

From sampled student tasks:

ExposureIndex = S1+S2+S3+S4/4

Where:

- S1 = % higher-order tasks attempted
- S2 = presence of abstraction prompts
- S3 = revision behaviour observed
- S4 = balanced cognitive demand

Range: 0–1

Step 3: Compute CoverageFactor

CoverageFactor = Students sampled/Total students in cohort

Capped at 1.

This prevents small samples from inflating readiness.

Step 4: Compute StudentReadiness

StudentReadiness0-8 = 8× (ExposureIndex × CoverageFactor)

Range: 0–8

Aggregate Ecosystem Readiness

EcosystemReadiness0-15 =ParentReadiness0-7 + StudentReadiness0-8

Maximum = 15

Interpretation Framework

Pattern	Interpretation
High internal readiness + Low ecosystem	Early stage shift, external lag
Low internal + High ecosystem	Perception distortion
High both	Institutionalised shift
Low both	Structural rebuild needed

Critical Safeguard

Ecosystem is weighted 15 (not 25) because:

- It is downstream
- It is sample-sensitive
- It validates direction, not maturity

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